

Since you're starting your **FAQ Search Engine** project, understanding **N-Grams** is important because they help capture phrases instead of only individual words.

What is an N-Gram?

An **N-Gram** is a sequence of **N consecutive words** (or characters) from a piece of text.

- **N = 1** → Unigram
- **N = 2** → Bigram
- **N = 3** → Trigram
- **N = 4** → Four-gram

EXAMPLE SENTENCE

```
I love learning NLP
```

1. UNIGRAM (N = 1)

Each individual word is one token.

```
I
love
learning
NLP
```

Output:

```
[I]
[love]
[learning]
[NLP]
```

2. BIGRAM (N = 2)

Take two consecutive words.

```
I love
love learning
learning NLP
```

Output:

```
[I love]
[love learning]
```

```
[learning NLP]
```

3. TRIGRAM (N = 3)

Take three consecutive words.

```
I love learning  
love learning NLP
```

Output:

```
[I love learning]  
[love learning NLP]
```

4. FOUR-GRAM (N = 4)

```
I love learning NLP
```

Output:

```
[I love learning NLP]
```

Another Example

Sentence:

```
How are you today
```

UNIGRAM

```
How  
are  
you  
today
```

BIGRAM

```
How are  
are you  
you today
```

TRIGRAM

```
How are you  
are you today
```

Formula

If a sentence has **W words**, then:

$$\text{Number of N-Grams} = W - N + 1$$

Example

Sentence:

I love learning NLP

Words = 4

UNIGRAM

$$4 - 1 + 1 = 4$$

BIGRAM

$$4 - 2 + 1 = 3$$

TRIGRAM

$$4 - 3 + 1 = 2$$

FOUR-GRAM

$$4 - 4 + 1 = 1$$

Why do we use N-Grams?

Without N-Grams, the sentence is treated as separate words.

Example:

I love machine learning

Bag of Words:

I
love
machine
learning

The relationship between words is lost.

With Bigrams:

I love
love machine
machine learning

Now the model knows that "**machine learning**" is a meaningful phrase.

Real-world Examples

SEARCH ENGINE

User searches:

machine learning tutorial

Bigrams:

machine learning
learning tutorial

Documents containing "**machine learning**" are ranked higher.

CHATBOT

User says:

reset password

Bigram:

reset password

Instead of treating:

reset
password

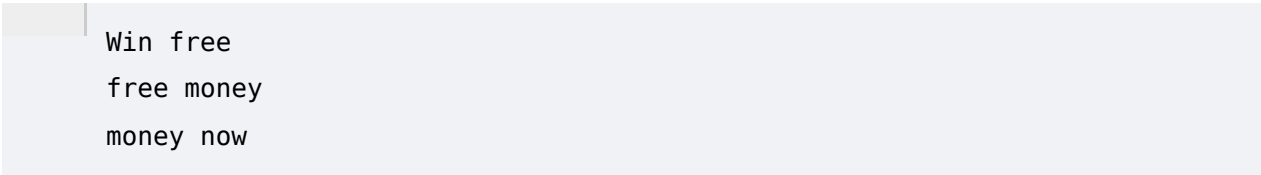
separately.

SPAM DETECTION

Sentence:

Win free money now

Bigrams:

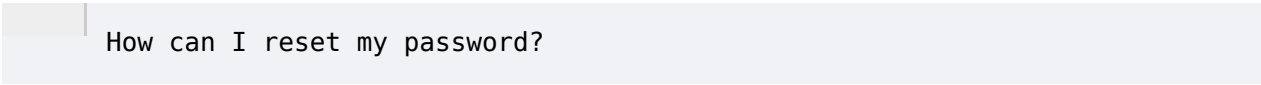


Win free
free money
money now

The phrase "**free money**" is a much stronger spam indicator than the individual words.

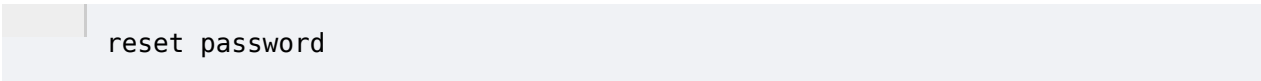
FAQ SEARCH ENGINE

FAQ:



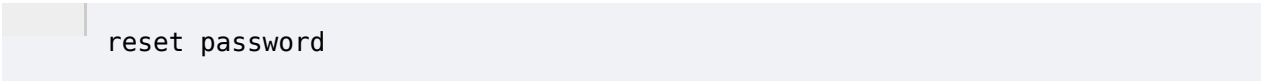
How can I reset my password?

After preprocessing:



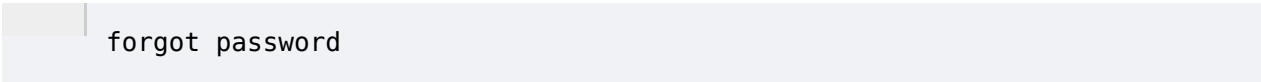
reset password

Bigram:



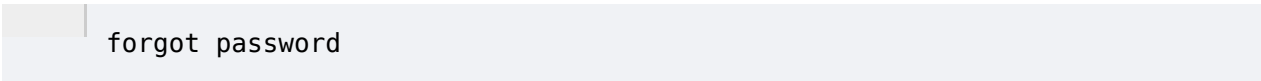
reset password

User query:



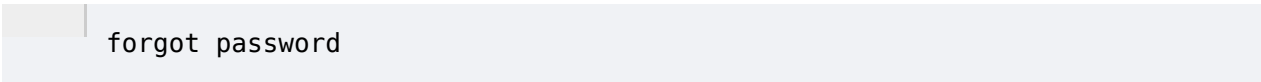
forgot password

After preprocessing:



forgot password

Bigram:



forgot password

Even if the exact words differ, using phrases (and later techniques like TF-IDF or semantic search) often gives more relevant matches than single words alone.

Advantages

- Captures word order.
 - Preserves short phrases.
 - Improves search quality.
 - Better for autocomplete.
 - Helps spam detection.
 - Useful for chatbots and FAQ search.
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Disadvantages

- Creates more features than unigrams.
 - Requires more memory.
 - Longer N (e.g., 4-grams or 5-grams) becomes sparse because exact phrases occur less frequently.
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C# Example

```
string sentence = "I love learning NLP";

List<string> words = sentence.Split(' ').ToList();

List<string> bigrams = new();

for (int i = 0; i < words.Count - 1; i++)
{
    bigrams.Add($"{words[i]} {words[i + 1]}");
}

foreach (var gram in bigrams)
{
    Console.WriteLine(gram);
}
```

Output:

```
I love
love learning
learning NLP
```

WHERE DOES N-GRAMS FIT IN THE NLP PIPELINE?

```
Raw Text
  ↓
Normalization
  ↓
Tokenization
  ↓
Stop Word Removal
  ↓
```

Stemming/Lemmatization



N-Grams



Bag of Words



TF



IDF



TF-IDF



Cosine Similarity

For your **FAQ Search Engine**, a practical approach is to generate **both unigrams and bigrams**. Unigrams help match individual keywords (like `password`), while bigrams capture meaningful phrases (like `reset password`), leading to more accurate search results. Later, you can feed these n-grams into your Bag of Words and TF-IDF calculations.